

To: KAUST Photonics Summer Camp 2026 Admissions Committee
From: Jamshidbek Olimov, Fergana State Technical University
Subject: Statement of Purpose for Photonics Summer Camp 2026

Dear Members of the Admissions Committee,

I am writing to express my strong enthusiasm for participating in the Photonics Summer Camp 2026 at King Abdullah University of Science and Technology (KAUST). As a 3rd-year Electrical Engineering student at Fergana State Technical University with a cumulative GPA of 3.88/4.00, I have developed a solid foundation in automatic control theory, electrical machines, and sustainable energy systems. I am eager to explore how advanced automation and sensor integration can intersect with modern photonics to solve global energy challenges.

During my academic journey, I have actively sought opportunities to apply theoretical knowledge to practical engineering problems. As the Team Leader of Scientific Projects at the EcoTech Club, I led multidisciplinary student teams in designing innovative environmental and engineering solutions. Furthermore, in my "Smart Trash Can" startup project, I worked as a hardware and software developer, integrating sensors and programming microcontrollers to build an automated waste management prototype with optimized low power consumption. These experiences solidified my skills in circuit design, sensor integration, and data processing using Python.

My ultimate research goal is to contribute to the development of highly efficient smart grid systems and sustainable energy automation. I believe that the future of energy efficiency lies in the bridge between classical electrical engineering and photonics, particularly in utilizing advanced optical sensors for real-time fault detection and grid optimization. I am particularly motivated to explore how photonic technologies can enhance energy efficiency and enable smarter infrastructure in developing regions like Uzbekistan.

I am particularly interested in the research of Dr. Nazek El-Atab on smart systems and miniaturized devices, which aligns closely with my background in sensor integration. Additionally, the work of Dr. Boon Ooi in semiconductor photonics and Prof. Andrea Fratalocchi in advanced photonic systems represents an exciting opportunity to deepen my understanding of modern optical technologies.

Participating in the KAUST Photonics Summer Camp will not only allow me to gain invaluable hands-on research experience in world-class laboratories but also enable me to experience KAUST's unique multicultural and collaborative environment. I am confident that my technical skills, leadership experience, and strong academic drive make me a well-suited candidate for this program.

Thank you for considering my application. I look forward to the possibility of contributing to and learning from the vibrant research community at KAUST.

Sincerely,

Jamshidbek Olimov
Electrical Engineering Student
Fergana State Technical University
jamshidbek.olimov@gmail.com
+998-90-777-04-04